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CMU Capstone Project Virtual Gold | Intro Meeting

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Attachments CMU Capstone Project Virtual Gold | Intro Meeting

Summary

Project scope discussion established hybrid model requirements and functional prototype objectives for secure enterprise AI integration.

Project Scope and Objectives
Participants established that the core deliverable is a functional hybrid system integrating local and cloud models to balance privacy and performance. The team must conduct discovery research to define a focus sector.

Technical Architecture Requirements
Mandatory project parameters include strict security measures and the utilization of open source models rather than building from scratch. Successful delivery requires demonstrating comparable performance between local and cloud-based systems.

Timeline and Implementation
The 15-week timeline necessitates immediate prototyping to test architectural assumptions during the discovery phase. Early experimentation will support the finalization of the project charter next week.

Decisions

Aligned

- **Mandatory hybrid architecture components and parameters** The team aligned on establishing a hybrid architecture requiring a local model that addresses privacy, security, and performance-to-cost trade-offs.
- **General-purpose AI model evaluation benchmarks** The group agreed that the hybrid model should be evaluated for general-purpose performance using established industry benchmarks.
- **Inclusion of local model enhancement** The participants aligned on including methods to enhance local model performance within the project scope.
- **Exclusion of graphical interface development** The group agreed that developing a graphical user interface is not required at the current stage, prioritizing underlying architecture instead.

Next steps

[The group] Research Hybrid Models: Utilize large language models to explore the current state of hybrid AI and relevant case studies.

[Anmol Sahu] Share Weekly Schedule: Send the course timeline and project milestones to the client team.

[The group] Finalize Project Charter: Define the project scope and complete the charter document.

[Urte Jesina] Schedule Weekly Meetings: Coordinate a recurring time for project updates with the student team.

[Urte Jesina] Establish Email Thread: Initiate a formal email thread to facilitate communication between the stakeholders.

[Anmol Sahu] Establish Communication Space: Create a shared space in Google Chat or Slack for ongoing team dialogue.

Details

- **Introduction and Project Context**: Participants from Virtual Gold (Inderpal Bhandari, Alex, and Urte Jesina) and the student team (Anmol Sahu, Yudi Guo, Karina, Mark, and Zhexuan) introduced themselves. Inderpal Bhandari explained the motivation for the capstone project: a hybrid AI platform using local processing as an orchestrator to maintain control over privacy and security, as current cloud-based models often conflict with enterprise needs to protect proprietary data.
- **Project Scope and Discovery**: Yudi Guo inquired about current practices and blockers, identifying data privacy and the conflict between local and cloud models as key issues. Inderpal Bhandari advised that the students should conduct their own discovery research using tools like ChatGPT or Claude to define the scope and state-of-the-art, emphasizing that the project is flexible and the team may choose to focus on specific sectors like financial or real estate data.
- **Project Deliverables**: Regarding deliverables, Yudi Guo asked for clarification on whether to focus on a prototype or a report. Inderpal Bhandari and Alex clarified that the goal is a functional hybrid system demonstrating the interaction between a local model and cloud models, with the requirements detailed in section three of the project description serving as the primary deliverable framework.
- **Performance Metrics and Obstacles**: Alex highlighted that the performance gap between local and cloud models is a critical challenge, and the team must demonstrate how to maintain privacy while achieving comparable performance. Inderpal Bhandari noted that cost is another primary driver for adopting a local model, and these measurable performance and cost factors will be essential to evaluate the project's success.
- **Meeting Cadence**: Yudi Guo asked about the meeting schedule. Inderpal Bhandari suggested meeting weekly during the initial discovery phase, with the potential to transition to a bi-weekly schedule once the project is better defined, keeping in mind the December project deadline.
- **Technical Parameters and Architecture**: Anmol Sahu inquired about compulsory parameters for the architecture. Inderpal Bhandari confirmed that a hybrid architecture with a local model is mandatory, as are strict privacy and security measures. Alex added that the team should benchmark the model against general-purpose AI standards to ensure performance.
- **Infrastructure and Sandbox**: Mark asked about infrastructure and model selection. Inderpal Bhandari and Alex explained that they would provide a cloud-based sandbox for testing, though the final solution must be designed to run on locally attainable hardware.

- **Security and International Models**: Inderpal Bhandari and Alex addressed the use of international models, noting that while they may offer high performance, they will face increased scrutiny regarding security and potential malware. Alex suggested architectural modularity so that the team can swap between different model options if necessary.
- **Enhancement Methods and Scope**: Anmol Sahu asked if enhancing local models is within scope. Inderpal Bhandari confirmed that using and enhancing existing open-weight or open-source models is the expected approach, as building a new model from scratch is not viable within the project timeline.
- **Data Availability**: Anmol Sahu inquired about data availability. Inderpal Bhandari stated that the team would provide necessary data once the students finalize their chosen use case and scope.
- **System Requirements and Latency**: Karina asked about latency, accuracy, and platform expectations. Inderpal Bhandari and Alex emphasized that the local-hybrid system must provide performance comparable to cloud models, and while a graphical interface is not a priority, they should focus on the underlying architecture.
- **Project Timeline**: Anmol Sahu outlined the 15-week project timeline. Inderpal Bhandari noted that the team is currently on track, with the goal of finalizing the project charter by the following week.
- **Prototyping Strategy**: Alex suggested incorporating prototyping earlier into the discovery phase to test architectural assumptions. Inderpal Bhandari agreed, noting that early experimentation is necessary to support the project's analysis.
- **Communication Logistics**: The group discussed communication logistics. Urte Jesina agreed to coordinate meeting times, and the group decided to use a dedicated space, such as Google chat, to facilitate ongoing communication.

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